AARUSHI SINGH

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Education

Manipal University Jaipur, B.Tech Computer Science Engineering with specialization in Artificial Intelligence and Machine Learning — Sept 2022 – Sept 2026

- GPA: CGPA : 8.42 (till Sem V)

Delhi Public School Indirapuram, PCM with Psychology — April 2008 – April 2022

- Percentage : 93.4

Skills

- **Programming Languages:** Python, Go, JavaScript, SQL, C/C++
- **Web and Backend:** Flask, Gin-Gonic (Go), GORM, Swagger, Twilio API
- **Machine Learning and AI:** Scikit-learn, TensorFlow, Keras, OpenCV, MediaPipe, NumPy, Pandas
- **Deep Learning Models:** LSTM, ARIMA, SRGAN, GANs, Zero-DCE, K-Means/K-Medoids, LexRank, SVD
- **Tools and Platforms:** Tableau(certified), Git, Microsoft, Google Colab, Jupyter Notebooks, VS Code
- **Research and Evaluation:** ROUGE Metrics, Time Series Forecasting, Image Super-Resolution, Low-Light Enhancement
- **Soft Skills and Roles:** Technical writing, Event coordination, Cross-cultural collaboration, Team leadership

Projects

Tweepify: Twitter Campaign Manager
Developed Tweepify, an AI-powered Twitter campaign manager, achieving an 80% reduction in manual content creation and scheduling effort. Engineered a full-stack web application to generate tweets with 95% contextual relevance, boosting engagement by approximately 45%. Implemented an automated scheduling algorithm that reduced campaign setup time by 70% and improved visibility through optimal timing. Integrated real-time analytics with under 1-second latency using WebSocket, and ensured secure authentication via OAuth1.0a and Firebase. Overall, enabled users to plan and launch campaigns 4x faster through an intuitive, template-driven interface.

Comparative Text Summarization System
Designed and implemented an unsupervised text summarization framework using four models—Baseline, LSA, LexRank, and K-Medoids—achieving a 42% increase in ROUGE-1 F1 score and 35% improvement in ROUGE-L recall over the baseline. Utilized TF-IDF, SVD, cosine similarity, and Jaccard distance for sentence selection, ensuring 100% independence from pretrained models. Automated summary evaluation reduced manual analysis time by over 50%, and performance visualization via Python dashboards enabled comparative insights across models to optimize contextual relevance and algorithmic efficiency.

Hand-Gesture-based Mouse Control
Engineered a gesture-controlled mouse system using a standard webcam, achieving 90% average gesture recognition accuracy across seven distinct actions including cursor movement, clicks, scroll, zoom, and screenshot. Utilized MediaPipe for 21-point hand landmark detection and computed joint angles to classify gestures using 100% rule-based logic (no training data or deep learning). Delivered real-time interaction with 25–30 FPS responsiveness and 150–300 ms latency, while reducing jitter by 60% through movement smoothing. System functioned reliably across >5 lighting conditions and hand distances (30–60 cm), with <500MB memory usage, enabling deployment on mid-range laptops without GPU acceleration.

MediTrack - Integrated Medicine Adherence and Management Platform (MUJ HackX)
Implemented an integrated medicine adherence and management platform platform providing users with a complete solution for managing their healthcare needs by integrating features like medicine adherence reminders via WhatsApp, powered by the Twilio API. The platform ensures the security of its frontend components by rendering them on the backend. Additionally, user data is stored in CSV format, enabling data science applications like disease risk prediction and analysis. Also created dashboards in Tableau to analyze disease trends in MediTrack dataset.

Gin-based CRUD API Development using GORM in Go
Built a CRUD API using GORM and Go, focusing on HTTP request handling through the Gin framework. This project involved setting up a scalable and efficient system for handling API endpoints. I learned how to write custom logger functions to improve error tracking and implemented Swagger annotations for better API documentation. Error handling and input validation were integral aspects of the project to ensure the reliability and integrity of the API's operations.

Experience

CS Intern, Sinch – Noida — July 2024 – Aug 2024

- Implemented Swagger documentation for the company's campaign creation code, enhancing development team efficiency.
- Developed and integrated a custom logger into the campaign delivery back-end code, improving client delivery processes.

Current Research

Low-Light Object Detection Using Zero-DCE and Super-Resolution GANs

- Researching methods to enhance low-light images using Zero-DCE, alongside exploring Super-Resolution GANs to improve image resolution focusing on optimizing model performance and improving image quality in challenging lighting conditions.